Describe the parts, features, blood supply and histology of stomach

Stomach is a muscular elastic bag forming widest part of GIT
Connected to oesophagus above and duodenum below.

Location: upper part of abdominal cavity occupying epigastric and umbilical region

Shape: J shaped

Length:25 cm

Capacity: 1500 to 2000ml, Birth: 30ml

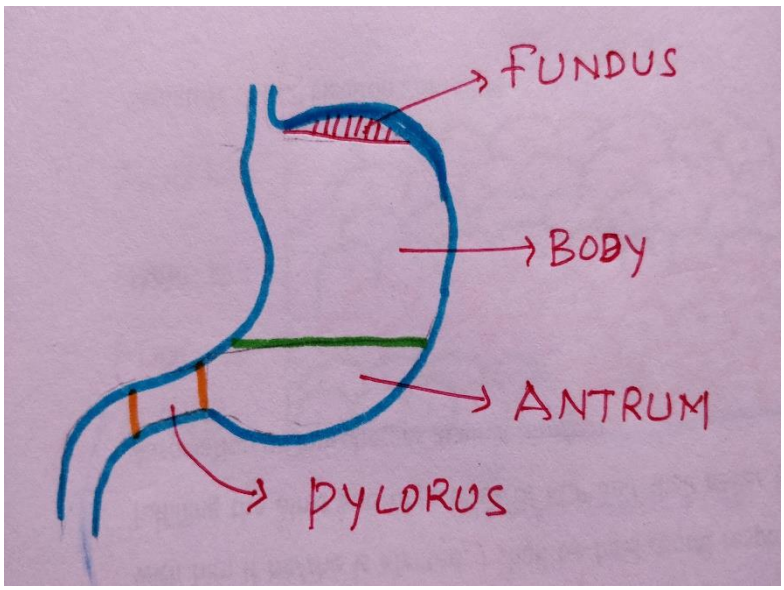
Sphinchters: cardiac sphincter and pyloric sphincter

Curvatures: lesser curvature and greater curvature

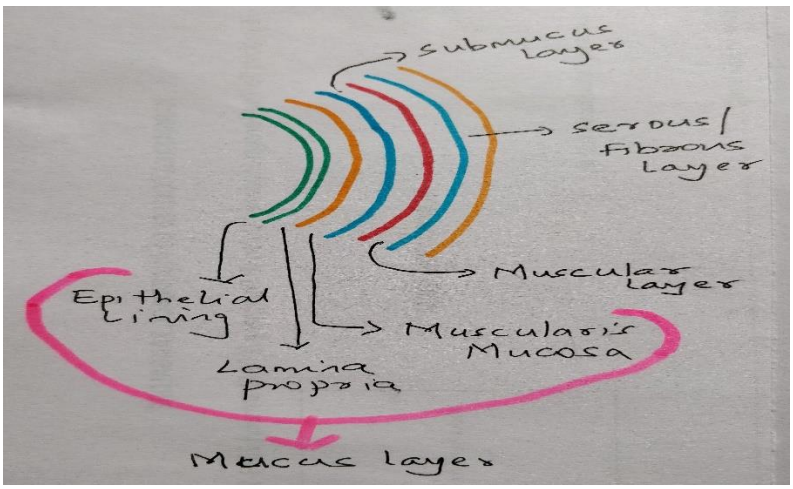
Surfaces: anterio superior surface and posterior inferior surface

Parts of stomach: stomach is composed of 4 parts Fundus, Body, Antrum and Pylorus

At the distal end of the pyloric antrum is the pyloric sphincter, guarding the opening between the stomach and duodenum. When the stomach is inactive the pyloric sphincter is relaxed and open, when stomach contains food the sphincter is closed



Layers of Stomach:



Stomach is composed of 4 layers 1. Mucus layer 2. Sub mucus layer 3. Muscular and 4. Serous Layer

1	Mucus layer	Its composed of 3 lining i. Columnar epithelium ii. Lamina propria: connective tissue iii. Muscularis Mucosa: composed of smooth muscle fibers
2	submucus layer	It contains loose collagen fibers, elastic fibers, reticular fibers and connective tissue, blood vessels. It also contains nerve plexus and sub mucosal plexus.
3	muscular layer	Composed of smooth muscle fibers. Its again divided into 3 layers a. Inner oblique layer b. Middle circular layer c. Outer longitudinal layer
4	serous layer	Serous layer is formed by connective tissue of meso epithelial cells

Nerve innervations:

Parasympathetic: vagus nerve

Sympathetic: splanchnic nerve

Histology: Every sq mm of stomach contains 60 to 100 pits or glands

Gastric Juice: About 2 liters of gastric juice are secreted daily by specialized Secretory glands in the Mucosa. It consists of

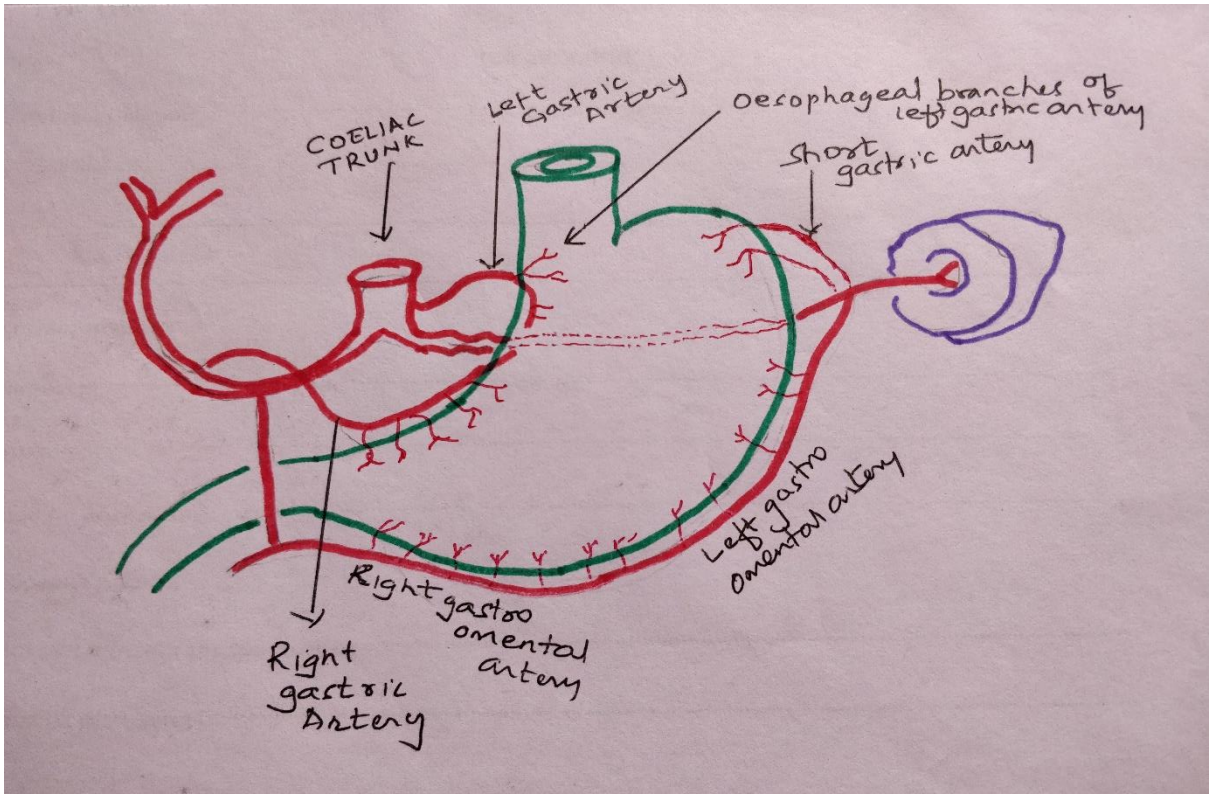
s.n	Secretory cells	Components	functions
1	Surface Mucous Cells	Mucus	Protects inner lining of stomach
2	Neck Mucous Cells	Mucus	Protects inner lining of stomach
3	Chief Cells	Pepsinogen	It's the inactive form of pepsin. In the presence of Hcl Pepsinogen is converted to pepsin Pepsin breakdowns proteins into peptides
		Gastric Lipase	Breakdowns triglyceride into fatty acid and glycerol
		Rennin	Curdling of milk
		Gastric Amylase	Breakdowns starch to dextrin and maltase
		Gelantinase	Breakdowns gelatin and collagen to peptide

		Urase.	Breakdowns urea to ammonia
4	Parietal Cells	Hydrochloric acid	1. it acts as primary defense mechanism by killing microorganism entering into GIT 2. converts inactive pepsinogen to pepsin 3. it promotes the denaturation of proteins there facilitate the digestion of protein
		Intrinsic Factor	Essential for the digestion of vitamin B12
5	G Cells	Gastrin	Promotes the secretion of gastric acid(Hcl) by the parietal cells of the stomach
6	Water		Dilutes food and other substances, facilitates mixing

<u>Blood supply to stomach</u>: blood supply to stomach is done by coeliac trunk which is the branch of aorta.

Left gastric artery(branch of coeliac trunk): provides blood supply to cardiac area of stomach

Short gastric artery(branch of splenic artery): supply blood to fundus of stomach



Right gastric artery(branch of hepatic artery): supply blood to lesser curvature of stomach

Right gastro omental artery(branch of hepatic artery): supply blood to greater curvature of stomach

Left gastro omental artery(branch of splenic artery): supply blood to greater curvature of stomach